

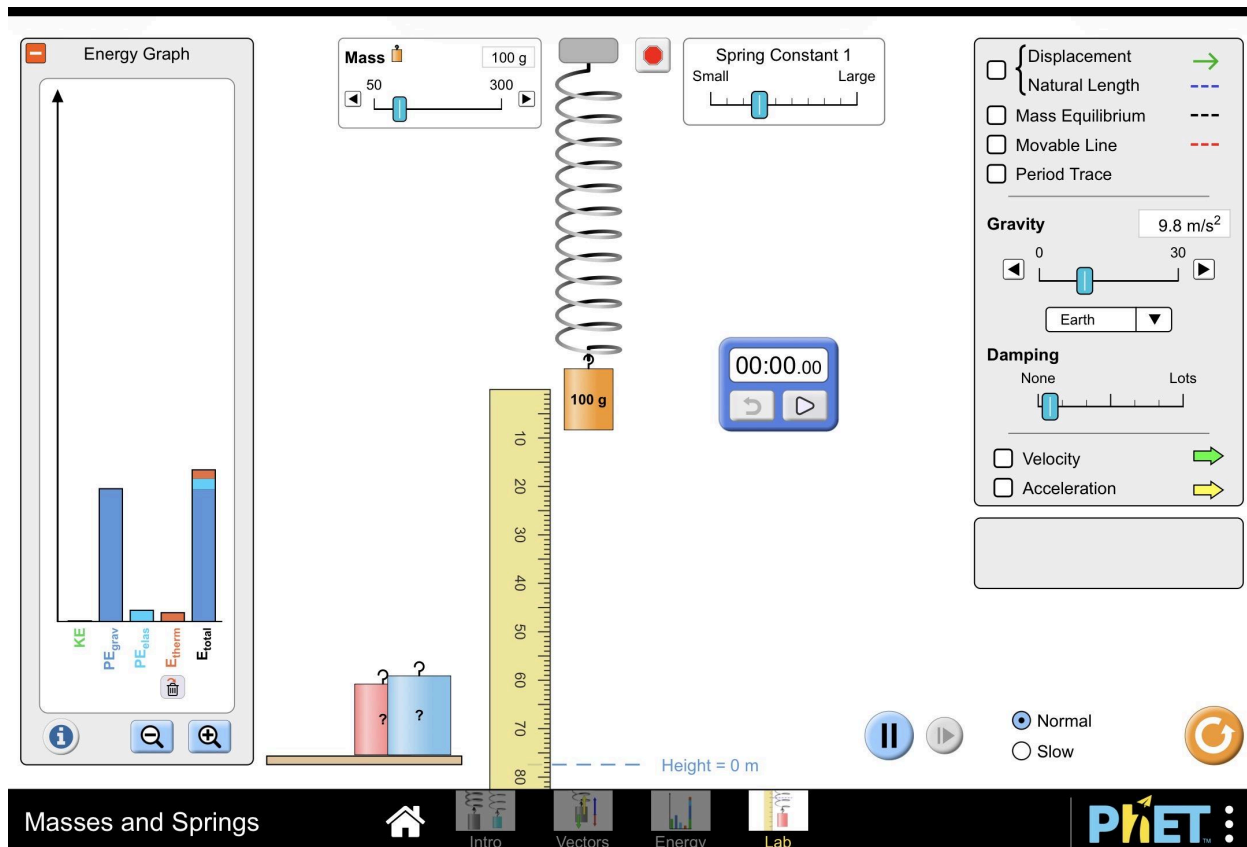
Steven Perez
Jacob Hobensack
Kaiden Fraser
Bear Ramsay

Unit 10 Lab Report

Purpose:

to determine the graphical and mathematical relationship between mass and change of starting position on period

Apparatus:



Procedure:

1. Open the "Masses and Springs" digital lab and select the "Slow" speed setting.
2. Place the orange mass on the spring and press the red stop button to find the equilibrium position
3. Align the 0 cm mark of the ruler with the bottom of the mass while it is at rest
4. Pause the simulation and pull the mass down to a fixed starting point
5. Press the play button on the timer, and press play button for the lab
6. Pause the simulation at the exact moment the mass returns to its lowest point after one full cycle. Record the time shown on the stopwatch in a data table
7. Use the slider to increase the mass by 50g and repeat the timing process for 8 to 10 trials
8. Reset the lab and keep the mass constant for the next set of tests
9. Change the starting displacement by 5cm increments starting at 5 cm.
10. Record the time for one oscillation at each new starting height for 8 to 10 trials in a table.

Data:

	Spring Lab ...			
	Mass (g) ...	Period (M) (s) ...	Distance (cm) ...	Period (x) (s) ...
1	100	0.83	5	0.85
2	125	0.94	10	0.85
3	150	1.01	15	0.85
4	175	1.10	20	0.85
5	200	1.17	25	0.85
6	225	1.24	30	0.85
7	250	1.30	35	0.85
8	275	1.36	40	0.85
9	300	1.43	45	0.85

Analysis:

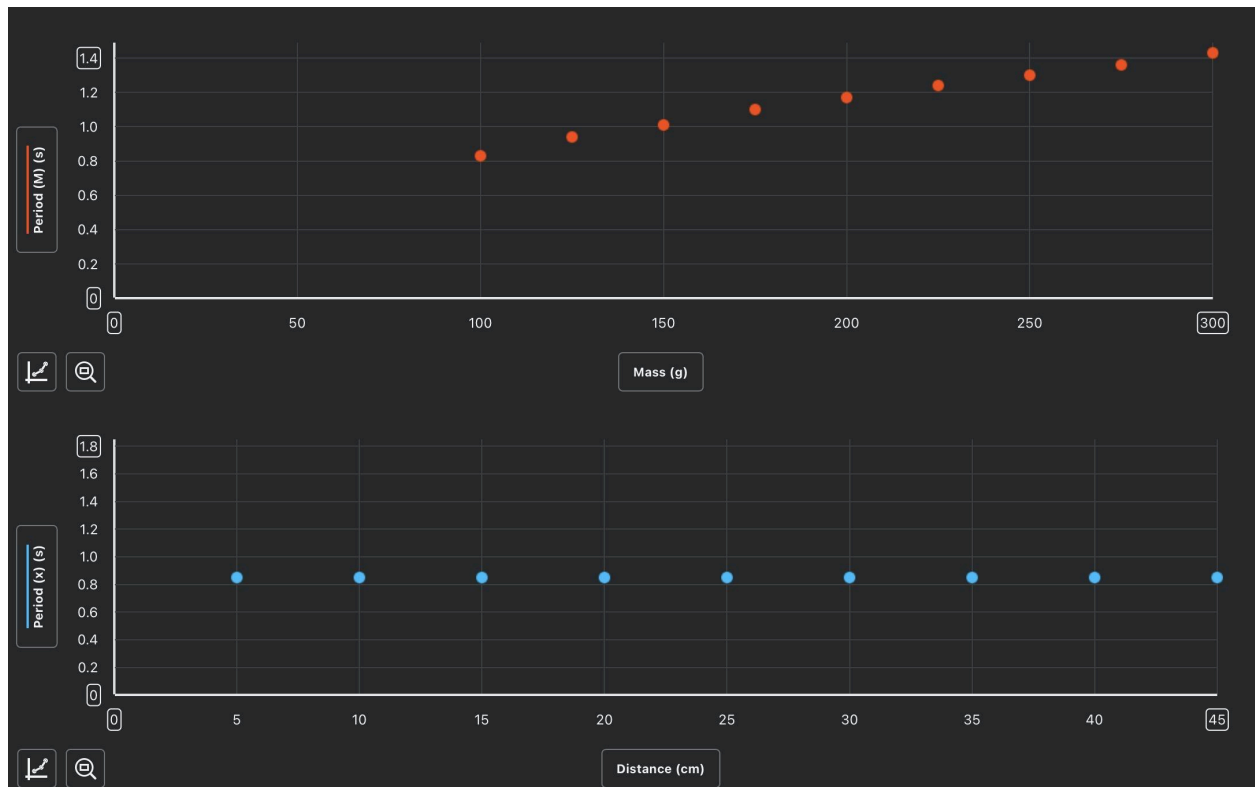
	Spring Lab					...
	Mass (g) ...	Period (M) (s) ...	Period^2 (... (s^2) ...	Distance (cm) ...	Period (x) (s) ...	...
1	100	0.83	0.689	5	0.85	
2	125	0.94	0.884	10	0.85	
3	150	1.01	1.020	15	0.85	
4	175	1.10	1.210	20	0.85	
5	200	1.17	1.369	25	0.85	
6	225	1.24	1.538	30	0.85	
7	250	1.30	1.690	35	0.85	
8	275	1.36	1.850	40	0.85	
9	300	1.43	2.045	45	0.85	

$$5 / (2.045) = 0.10225$$

$$.10225 > 0.3404$$

Period	Period ²
0.83	.689
0.94	.884
1.01	1.020
1.10	1.210
1.17	1.369
1.24	1.536
1.30	1.690
1.36	1.850
1.43	2.045

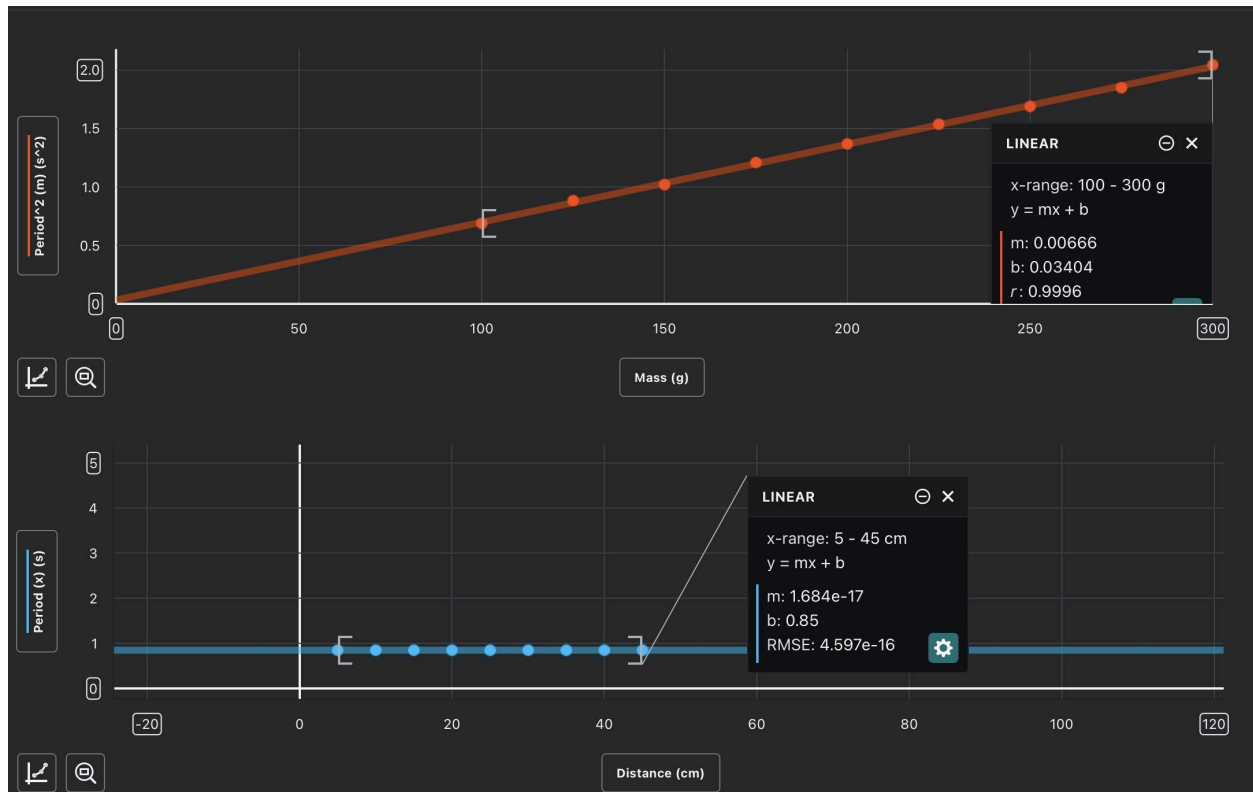
Original Mass/Distance stretched vs. Period graphs



Linearized Mass/Distance stretched vs Period graphs

$$T^2 = (0.00666 \text{ s}^2/\text{g})m + (0.03404 \text{ s}^2)$$

$$T = 0.85 \text{ s}$$



Due to the 5% rule, the intercept in the Mass vs. Period² graph goes away since 0.03404 is less than 5% of the greatest Period² value of 2.045.

$$T^2 = (0.00666 \text{ s}^2/\text{g})m$$

Conclusion:

The Mass vs. Period graph displayed a parabolic curve. In order to linearize it, period was squared, to produce a mass vs. period² graph. Once linearized, a linear regression was applied, yielding the equation $T^2 = (0.00666 \text{ s}^2/\text{g})m + (0.03404 \text{ s}^2)$. Applying the 5% rule, since 0.03404 s² is less than 5% of the greatest Period² value of 2.045 s², the y-intercept was removed, making the equation $T^2 = (0.00666 \text{ s}^2/\text{g})m$. Taking the square root of both sides gives the equation of the original curve $T = \sqrt{(0.00666 \text{ s}^2/\text{g})m}$. The slope of 0.00666 s²/g represents how much the square of the period increases per gram of mass added to the spring. In the generalized form of the equation, the equation is $T = \sqrt{(am)}$, where T is the period, m is the mass, and a is a constant. For the distance vs. Period graph, the data showed a horizontal line at T=0.85s, showing that changing the starting displacement had no effect on the period of oscillation. One source of error would be pausing the simulation at the exact moment the hanging mass completed one full cycle, which could cause small timing inaccuracy. This can be fixed by running multiple trials of each mass and taking the average to reduce any mistimed measurements.